Scalable Generalized Linear Models (GLM)

XX-XX-XX

Shuo Zhou, PhD

COM6012 Scalable Machine Learning, Spring 2026

University of Sheffield

A World TOP 100 University

Content

XX-XX-XX

- Linear regression, logistic regression, and Poisson regression
- The exponential family
- Generalized linear models (GLM)
- Iteratively reweighted least squares (IRLS)
- GLM in Spark ML

Week 4 Objectives

XX-XX-XX

- Relate linear, logistic, Poisson and Gamma regression to the unified ***generalized linear models (GLM)*** framework.
- Derive the ***exponential family*** form of **Gaussian**, **Bernoulli**, **Poisson**, and **Gamma** distributions.
- Identify the **natural parameter**, **mean parameter**, and **link function** in a GLM and explain how they are connected.
- Derive the GLM with ***canonical link functions*** (identity, logit, log).
- Explain ***iteratively reweighted least squares (IRLS)*** for GLM optimization.
- Choose and implement appropriate GLM in PySpark for different data types and learning tasks.

Content

XX-XX-XX

- **Linear regression, logistic regression, and Poisson regression**
- The exponential family
- Generalized linear models (GLM)
- Iteratively reweighted least squares (IRLS)
- GLM in Spark ML

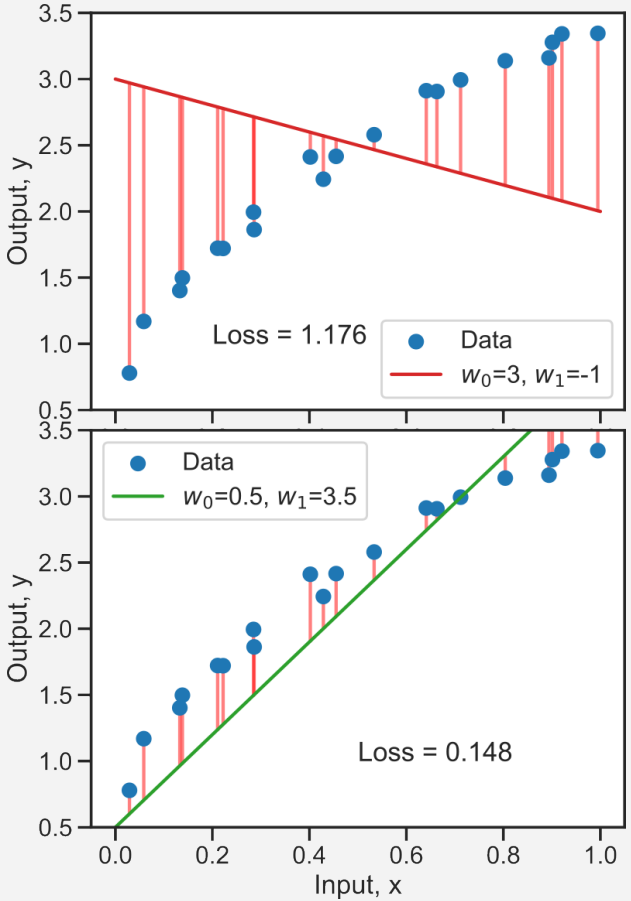
Linear regression

Recall in COM6509 MLAI

Mean Squared Error (MSE):

$$E = \frac{1}{N} \sum_{i}^{N} (y_i - f(\mathbf{x}_i, \mathbf{w}))^2$$

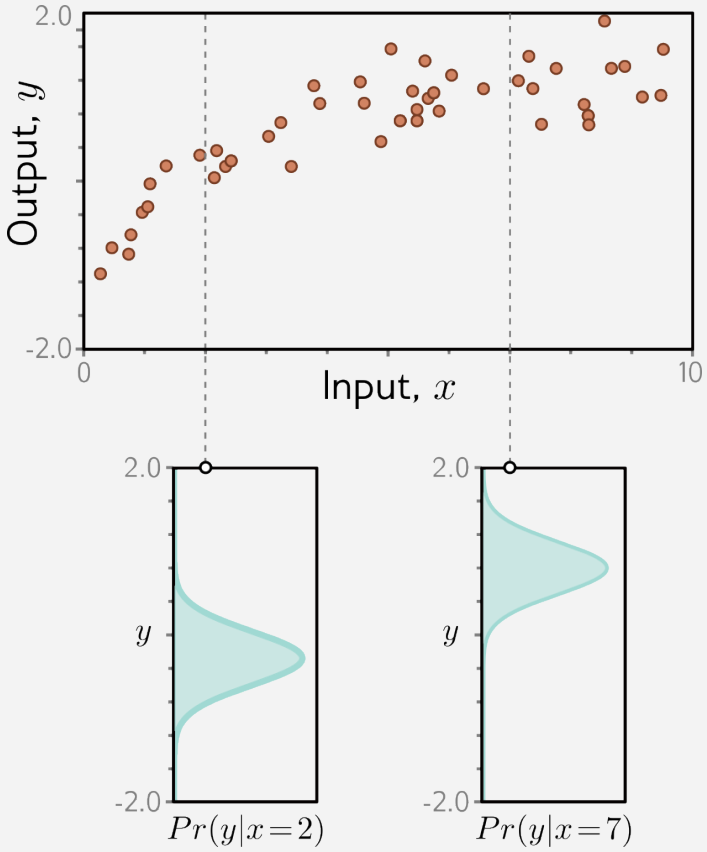
Where $(\mathbf{x}_i, y_i)$ is a training data point and $f(\mathbf{x}_i, \mathbf{w})$ is our prediction model parameterised by the weight vector.



Linear regression

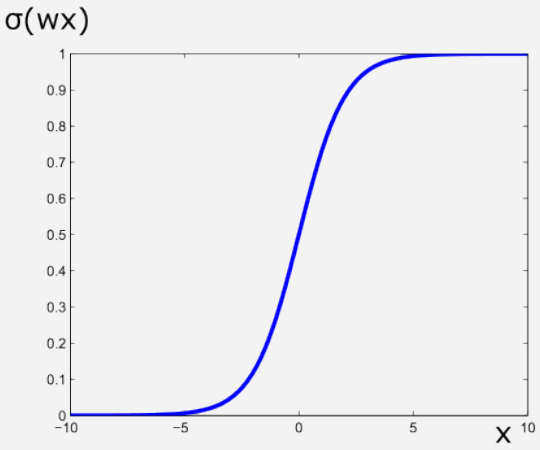
$$p(y_i|\mathbf{x}_i, \mathbf{w}, \sigma^2) = \mathcal{N}(y_i|\mathbf{w}^\top \mathbf{x}_i, \sigma^2)$$

$$\mathbb{E}(y_i|\mathbf{x}_i) = \mathbf{w}^\top \mathbf{x}_i, y \in \mathbb{R}$$



Logistic regression

Recall last week



$$p(y_i|\mathbf{x}_i, \mathbf{w}) = \mathrm{Ber}(y_i|\sigma(\mathbf{w}^\top \mathbf{x}_i))$$

$$p(Y = 1) = \sigma(z) = \frac{1}{1 + \exp(-z)} = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x})}$$

$$\max_{\mathbf{w}} \prod_{i=1}^{N} p(y_i|\mathbf{w}, \mathbf{x}_i) \to \min_{\mathbf{w}} \sum_{i=1}^{N} -\log p(y_i|\mathbf{w}, \mathbf{x}_i)$$

$$\mathbb{E}(y_i|\mathbf{x}_i) = \sigma(\mathbf{w}^\top \mathbf{x}_i) = \frac{\exp(\mathbf{w}^\top \mathbf{x}_i)}{1 + \exp(\mathbf{w}^\top \mathbf{x}_i)}, y_i \in \{0, 1\}.$$

Count data

- Non-negative integer values, e.g. number of road casualties, $Y \in \mathbb{N} = 0, 1, 2, 3, \ldots$

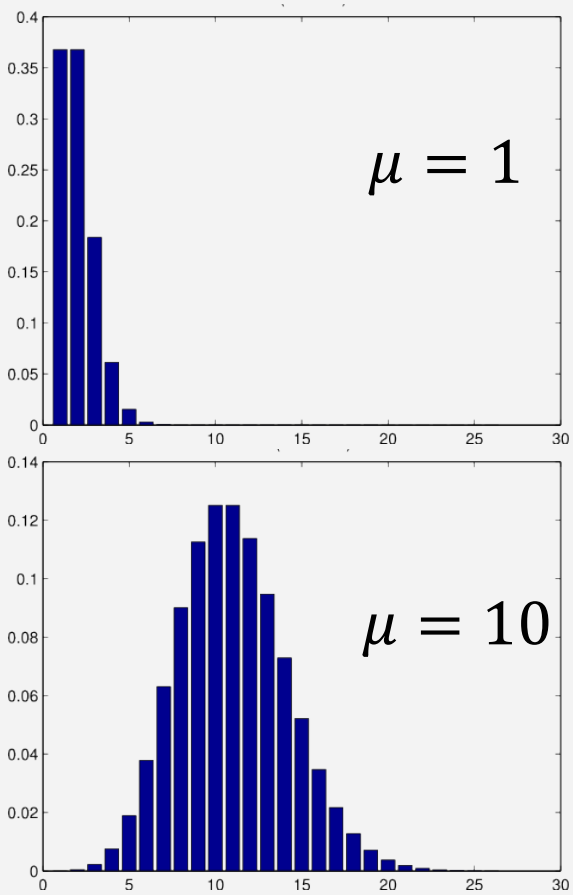
- **Discussion**: Can we use linear regression or logistic regression for count data? Why?

Poisson distribution

- A discrete probability distribution and the probability mass function (PMF) is

$$\mathrm{Poi}(y|\mu) = \exp(-\mu)\frac{\mu^y}{y!},$$

where $\mu > 0$ is called the *rate parameter*, and $y \in \{0, 1, 2, \ldots\}$



Poisson regression

- Definition: $p(y_i|\mathbf{x}_i, \mathbf{w}) = \mathrm{Poi}(y_i|\exp(\mathbf{w}^\top \mathbf{x}_i))$,

  where $\mathrm{Poi}(y|\mu) = \exp(-\mu)\frac{\mu^y}{y!}$

- The log probability density function (PDF):

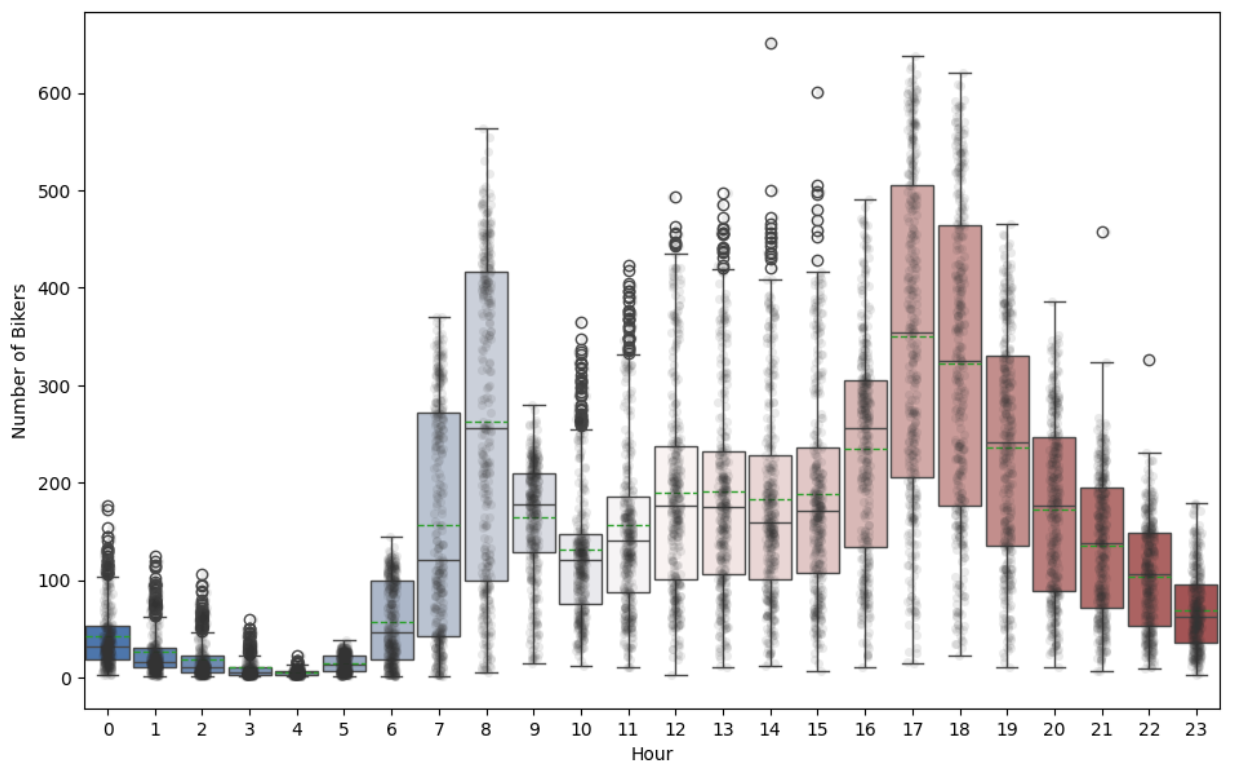
$$\log p(y_i|\mathbf{x}_i, \mathbf{w}) = y_i \log \mu_i - \mu_i - \log(y_i!),$$

  where $\mu_i = \mathbb{E}(y_i|\mathbf{x}_i) = \exp(\mathbf{w}^\top \mathbf{x}_i)$

- Likelihood

$$p(y_1, y_2, \ldots, y_N|\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N; \mathbf{w}) = \prod_{i=1}^{N} \exp(-\exp(\mathbf{w}^\top \mathbf{x}_i))\frac{\exp(y_i \mathbf{w}^\top \mathbf{x}_i)}{y_i!}$$

Example: predicting the number of bike users



Source: https://pykale.github.io/transparentML/08-glm-svm/glm.html#generalised-linear-models

Generalized form?

- Linear regression $\mathbb{E}(y|\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$
- Logistic regression $\mathbb{E}(y|\mathbf{x}) = \dfrac{\exp(\mathbf{w}^\top \mathbf{x})}{1 + \exp(\mathbf{w}^\top \mathbf{x})}$
- Poisson regression $\mathbb{E}(y|\mathbf{x}) = \exp(\mathbf{w}^\top \mathbf{x})$
- General form? $g(\mathbb{E}(y|\mathbf{x})) = \mathbf{w}^\top \mathbf{x}$, or $\mathbb{E}(y|\mathbf{x}) = g^{-1}(\mathbf{w}^\top \mathbf{x})$

Content

XX-XX-XX

- Linear regression, logistic regression, and Poisson regression
- **The exponential family**
- Generalized linear models (GLM)
- Iteratively reweighted least squares (IRLS)
- GLM in Spark ML

Introduction

- By now you are familiar with different types of probability distributions: the **Gaussian**, the **Bernoulli**, the **Poisson**, etc.

- These are members of a broader class of distributions known as the **exponential family**.

Why the exponential family is important?

- It can be shown that the exponential family is the only family of distributions with finite-sized sufficient statistics.
- The exponential family is the only family of distributions for which conjugate priors exist.
- The exponential family can be shown to be the family of distributions that makes the least set of assumptions subject to some user-chosen constraints.
- The exponential family is at the core of generalized linear models.

Definition

A probability density function (PDF) or a probability mass function (PMF) $p(\mathbf{y}|\,\boldsymbol{\theta})$, with $\mathbf{y} \in \mathbb{R}^p$ and $\boldsymbol{\theta} \in \mathbb{R}^d$, is in the exponential family if it is of the form

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} h(\mathbf{y}) \exp\left[\boldsymbol{\theta}^\top \boldsymbol{\mathcal{T}}(\mathbf{y})\right],$$

where $Z(\boldsymbol{\theta}) = \int_y h(\mathbf{y}) \exp\left[\boldsymbol{\theta}^\top \boldsymbol{\mathcal{T}}(\mathbf{y})\right] d\mathbf{y}$.

- $h(\mathbf{y})$ is a scaling constant, often 1.
- $\boldsymbol{\theta}$ are known as the **natural parameters** or **canonical parameters**.
- $\boldsymbol{\mathcal{T}}(\mathbf{y}) \in \mathbb{R}^d$ is called a vector of **sufficient statistics**.
- $Z(\boldsymbol{\theta})$ is known as the **partition function**.

Definition

Distributions in the exponential family can also be expressed as

$$p(\mathbf{y}|\boldsymbol{\theta}) = h(\mathbf{y}) \exp\left[\boldsymbol{\theta}^\top \boldsymbol{\mathcal{T}}(\mathbf{y}) - A(\boldsymbol{\theta})\right],$$

where $A(\boldsymbol{\theta}) = \log Z(\boldsymbol{\theta})$.

- $A(\boldsymbol{\theta})$ is called the **log partition function** or **cumulant function**.
- If $\boldsymbol{\mathcal{T}}(\mathbf{y}) = \mathbf{y}$, we say it is a **natural exponential family**.

Example: Bernoulli

For the Bernoulli distribution, $y \in \{0, 1\}$, and we have

$$\begin{aligned}
\mathrm{Ber}(y|\mu) &= \mu^y (1-\mu)^{1-y} \\
&= \exp\left[y \log \mu + (1-y)\log(1-\mu)\right] \\
&= \exp\left[y \log\left(\frac{\mu}{1-\mu}\right) + \log(1-\mu)\right], \\
&= h(y)\exp\left[\boldsymbol{\theta}^\top \boldsymbol{\mathcal{T}}(y) - A(\boldsymbol{\theta})\right],
\end{aligned}$$

where
- $\theta = \log\left(\frac{\mu}{1-\mu}\right)$, known as **log-odds ratio**
- $\boldsymbol{\mathcal{T}}(y) = y$
- $A(\theta) = -\log(1-\mu)$
- $h(y) = 1$

Example: univariate Gaussian distribution

The univariate Gaussian can be written in an exponential family form as

$$\begin{aligned}\mathcal{N}(y|\mu,\sigma^2) &= \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}}\exp\left[-\frac{1}{2\sigma^2}(y-\mu)^2\right] \\ &= \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}}\exp\left[-\frac{1}{2\sigma^2}y^2+\frac{\mu}{\sigma^2}y-\frac{1}{2\sigma^2}\mu^2\right] \\ &= \frac{1}{Z(\boldsymbol{\theta})}\exp\left(\boldsymbol{\theta}^\top\boldsymbol{\mathcal{T}}(y)\right),\end{aligned}$$

where
- $\boldsymbol{\theta} = \begin{bmatrix}\mu/\sigma^2 \\ -1/2\sigma^2\end{bmatrix}$, $\boldsymbol{\mathcal{T}}(y) = \begin{bmatrix} y \\ y^2\end{bmatrix}$, $Z(\boldsymbol{\theta}) = \sqrt{2\pi}\sigma\exp\left\{\frac{\mu^2}{2\sigma^2}\right\}$
- $h(y) = 1$

Example: Poisson distribution

As a member of the exponential family, $\mathrm{Poi}(y|\mu) = \exp(-\mu)\frac{\mu^y}{y!}$ can be written as

$$\mathrm{Poi}(y|\mu) = \frac{h(y)}{Z(\theta)}\exp(\theta y),$$

where $\theta = \log\mu,\ h(y) = \dfrac{1}{y!},$ and $Z(\theta) = \exp(\mu),\ A(\theta) = \mu.$

- Recall that the expected value of y is equal to $\mathbb{E}[y] = \mu$.
- Then, the mean parameter μ can be recovered from the canonical parameter using

$$\mu = \exp(\theta).$$

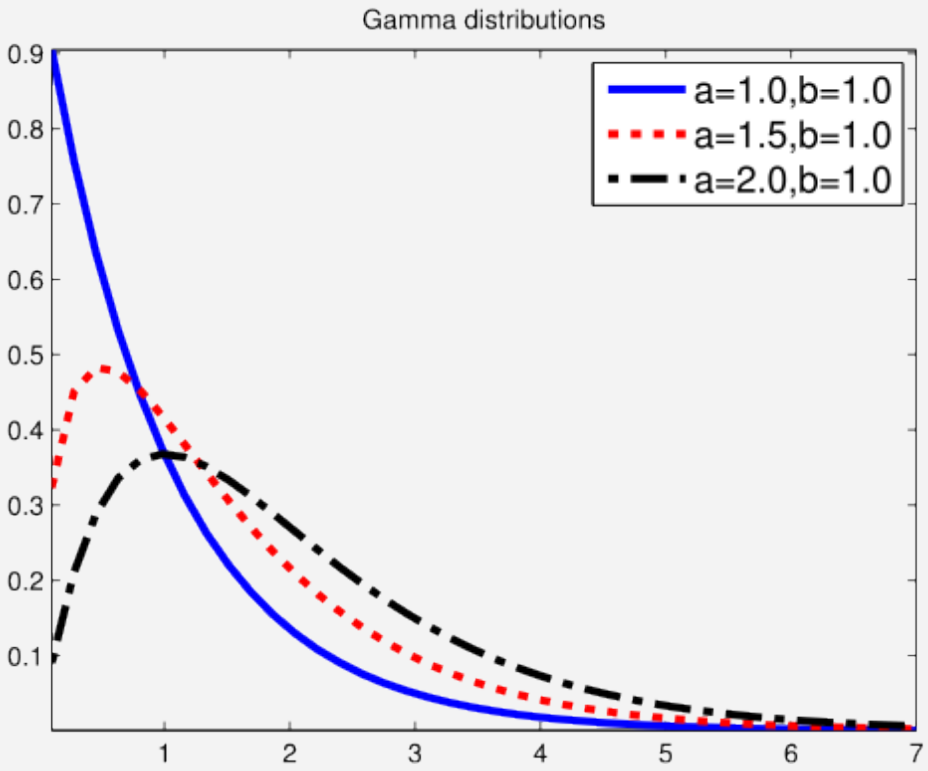
Example: Gamma distribution (I)

The Gamma distribution follows as

$$\mathrm{Ga}(y|a,b) = \frac{b^a}{\Gamma(a)} y^{a-1} \exp(-by),$$

where $a > 0$ (shape), and $b > 0$ (rate),

and $\Gamma(a) = \int_0^\infty u^{a-1} e^{-u} du$ is the Gamma function.



Example: Gamma distribution (II)

As a member of the exponential family, it can be written as

$$\mathrm{Ga}(y|a,b) = \frac{1}{Z(\boldsymbol{\theta})} \exp(\boldsymbol{\theta}^\top \boldsymbol{\mathcal{T}}(y)),$$

where
- $\boldsymbol{\theta} = \begin{bmatrix} a-1 \\ -b \end{bmatrix}$, $\boldsymbol{\mathcal{T}}(y) = \begin{bmatrix} \log y \\ y \end{bmatrix}$
- $Z(\boldsymbol{\theta}) = \dfrac{\Gamma(a)}{b^a}, \quad A(\boldsymbol{\theta}) = \log\Gamma(a) - a\log b$
- $h(y) = 1$

Recall the PDF of Gamma distribution: $\mathrm{Ga}(y|a,b) = \dfrac{b^a}{\Gamma(a)} y^{a-1} \exp(-by)$.

Content

XX-XX-XX

- Linear regression, logistic regression, and Poisson regression
- The exponential family
- **Generalized linear models (GLM)**
- Iteratively reweighted least squares (IRLS)
- GLM in Spark ML

Definition

- Linear, logistic, and Poisson regression are examples of generalized linear models (GLM).
- These are models whose output belongs to the exponential family.
- The mean parameters are a linear combination of the inputs, passed through a possibly nonlinear function, such as the logistic function.

GLM general form (I)

- We want to model the relationship between a response variable y_i, and an input vector $\mathbf{x}_i$.
- Let us first consider the case of an unconditional distribution for the response variable

$$p(y_i|\theta, \sigma^2) = \exp\left[\frac{y_i\theta - A(\theta)}{\sigma^2} + c(y_i, \sigma^2)\right],$$

  where θ is the natural parameter, $A(\cdot)$ is the log-partition function, σ^2 is the **dispersion parameter**, usually, $\sigma^2 = 1$, and c is the normalization constant.
- The expression for $p(y_i|\theta, \sigma^2)$ looks similar to the exponential family.

GLM general form (II)

- For example, in logistic regression, θ is the log-odds ratio

$$\theta = \log\left(\frac{\mu}{1-\mu}\right),$$

- where $\mu = \mathbb{E}[y] = p(y=1)$ is the mean parameter.
- To convert from the mean parameter μ to the natural parameter θ, we can use a function $\psi(\cdot), \theta = \psi(\mu)$.
- $\psi(\cdot)$ is uniquely determined by the form of the exponential family distribution.
- The mapping is **invertible**, so that $\mu = \psi^{-1}(\theta)$

Link function (I)

- Now let us add inputs or covariates.
- We first define a linear function of the inputs $\eta_i = \mathbf{w}^\top \mathbf{x}_i$.
- We now make the mean of the distribution be some invertible monotonic function of this linear combination.
- By convention, this function, known as the **mean function**, is denoted by $g^{-1}(\cdot)$, so

$$\mathbb{E}(y_i|\mathbf{x}_i) = \mu_i = g^{-1}(\eta_i) = g^{-1}(\mathbf{w}^\top \mathbf{x}_i).$$

- The inverse of the mean function, namely $g(\cdot)$, is called the **link function**.

Link function (II)

- We are free to choose almost any function we like for $g(\cdot)$, so long as it is invertible, and so long as $g^{-1}(\cdot)$ has the appropriate range.

- For example, in logistic regression, we set $\mu = g^{-1}(\eta_i) = \sigma(\eta_i)$

Relationships between functions

$$
\begin{tikzcd}
\boldsymbol{w} \arrow[dr] & & & & \\
& \eta_i \arrow[r, shift left, "g^{-1}"] & \mu_i \arrow[l, shift left, "g"] \arrow[r, shift left, "\Psi"] & \theta_i \arrow[l, shift left, "\Psi^{-1}"] \\
\boldsymbol{x}_i \arrow[ur] & & & &
\end{tikzcd}
$$

$g^{-1}(\cdot)$ is the **mean function**. $g(\cdot)$ is the **link function**.

GLM with canonical link function

- One particularly simple form of link function is to use $g = \psi$.
- This is called the **canonical link function**.
- In this case $\eta_i = \theta_i = \mathbf{w}^\top \mathbf{x}_i$, so the model becomes

$$p(y_i|\mathbf{x}_i, \mathbf{w}, \sigma^2) = \exp\left[\frac{y_i \mathbf{w}^\top \mathbf{x}_i - A(\mathbf{w}^\top \mathbf{x}_i)}{\sigma^2} + c(y_i, \sigma^2)\right].$$

Canonical link functions $g = \psi$ for some GLMs

Distribution	**Link** $g(\mu)$	$\theta = \psi(\mu)$	$\mu = \psi^{-1}(\theta)$
$\mathcal{N}(\mu, \sigma^2)$	identity	$\theta = \mu$	$\mu = \theta$
$\mathrm{Ber}(\mu)$	logit	$\theta = \log(\frac{\mu}{1-\mu})$	$\mu = \sigma(\theta)$
$\mathrm{Poi}(\mu)$	log	$\theta = \log(\mu)$	$\mu = \exp(\theta)$
$\mathrm{Ga}(a, b)$	inverse	$\theta = \mu^{-1}$	$\mu = \theta^{-1}$

Mean and variance of the response variable

- Recall $A(\cdot)$ as the ***cumulant function*** (or ***log partition function***)
- It can be shown that

$$\mathbb{E}[y|\mathbf{x}_i, \mathbf{w}, \sigma^2] = \mu_i = A'(\theta_i)$$
$$\mathrm{var}[y|\mathbf{x}_i, \mathbf{w}, \sigma^2] = \sigma_i^2 = A''(\theta_i)\sigma^2.$$

- Note the dispersion parameter σ^2 is not the variance of y.

Example: linear regression

In linear regression, the response variable $y_i \in \mathbb{R}$ follows a normal distribution,

$$
\begin{aligned}
p(y_i|\mu_i, \sigma^2) &= \mathcal{N}(y_i|\mu_i, \sigma^2) \\
&= \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} \exp\left[-\frac{1}{2\sigma^2}(y_i - \mu_i)^2\right] \\
&= \exp\left[\frac{y_i\mu_i - \frac{\mu_i^2}{2}}{\sigma^2} - \frac{y_i^2}{2\sigma^2} + \log\left(\frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}}\right)\right] \\
&= \exp\left[\frac{y_i\mu_i - \frac{\mu_i^2}{2}}{\sigma^2} - \frac{1}{2}\left(\frac{y_i^2}{\sigma^2} + \log(2\pi\sigma^2)\right)\right].
\end{aligned}
$$

GLM with canonical link functions: $p(y_i|\mathbf{x}_i, \mathbf{w}, \sigma^2) = \exp\left[\frac{y_i\mathbf{w}^\top\mathbf{x}_i - A(\mathbf{w}^\top\mathbf{x}_i)}{\sigma^2} + c(y_i, \sigma^2)\right].$

Example: linear regression

In linear regression, the response variable $y_i \in \mathbb{R}$ follows a normal distribution,

$$p(y_i|\mu_i, \sigma^2) = \exp\left[\frac{y_i\mu_i - \frac{\mu_i^2}{2}}{\sigma^2} - \frac{1}{2}\left(\frac{y_i^2}{\sigma^2} + \log(2\pi\sigma^2)\right)\right].$$

GLM with canonical link functions: $p(y_i|\mathbf{x}_i, \mathbf{w}, \sigma^2) = \exp\left[\frac{y_i\mathbf{w}^\top\mathbf{x}_i - A(\mathbf{w}^\top\mathbf{x}_i)}{\sigma^2} + c(y_i, \sigma^2)\right].$

The link function is $\theta_i = \mu_i = \mathbf{w}^\top\mathbf{x}_i$

$$A(\theta_i) = \mu_i^2/2,$$

$$\mathbb{E}[y_i] = \mu_i,$$

$$\mathrm{var}[y_i] = \sigma^2$$

Example: logistic regression

In logistic regression, the response variable follows a Bernoulli distribution

$$
\begin{aligned}
p(y_i|\mu_i) &= \mu_i^{y_i}(1-\mu_i)^{1-y_i} \\
&= \exp\left[\log\left(\frac{\mu_i}{1-\mu_i}\right) y_i - (-\log(1-\mu_i))\right].
\end{aligned}
$$

- The link function is the logit, $g(\mu_i) = \theta_i = \log\left(\frac{\mu_i}{1-\mu_i}\right) = \mathbf{w}^\top \mathbf{x}_i$.
- GLM with canonical link functions: $p(y_i|\mathbf{x}_i, \mathbf{w}, \sigma^2) = \exp\left[\frac{y_i \mathbf{w}^\top \mathbf{x}_i - A(\mathbf{w}^\top \mathbf{x}_i)}{\sigma^2} + c(y_i, \sigma^2)\right]$.
- $A(\theta_i) = -\log(1-\sigma(\theta_i))$, $\mathbb{E}[y_i] = \sigma(\theta_i)$, $\mathrm{var}[y_i] = \sigma(\theta_i)(1-\sigma(\theta_i))$.

Example: Poisson regression

In Poisson regression, the response variable follows a Poisson distribution

$$p(y_i|\mathbf{x}_i, \mathbf{w}, \sigma^2) = \exp[y_i \log(\mu_i) - \mu_i - \log(y_i!)].$$

- The link function is $\log$, $\log \mu_i = \mathbf{w}^\top \mathbf{x}_i$.
- GLM with canonical link functions: $p(y_i|\mathbf{x}_i, \mathbf{w}, \sigma^2) = \exp\left[\frac{y_i \mathbf{w}^\top \mathbf{x}_i - A(\mathbf{w}^\top \mathbf{x}_i)}{\sigma^2} + c(y_i, \sigma^2)\right].$
- $A(\theta_i) = \exp(\theta_i)$, $\mathbb{E}[y_i] = \exp(\theta_i) = \mu_i$.

Content

XX-XX-XX

- Linear regression, logistic regression, and Poisson regression
- The exponential family
- Generalized linear models (GLM)
- **Iteratively reweighted least squares (IRLS)**
- GLM in Spark ML

Log-likelihood for a GLM

- One of the appealing properties of GLMs is that they can be fit using the same methods that we used to fit logistic regression.
- In particular, the log-likelihood has the following form

$$\mathcal{L}(\mathbf{w}) = \frac{1}{\sigma^2}\sum_{i=1}^{N}\mathcal{L}_i = \frac{1}{\sigma^2}\sum_{i=1}^{N}[\theta_i y_i - A(\theta_i)],$$

where $\mathcal{L}_i = \theta_i y_i - A(\theta_i)$.

- Recall GLM:

$$p(y_i|\theta,\sigma^2) = \exp\left[\frac{y_i\theta - A(\theta)}{\sigma^2} + c(y_i,\sigma^2)\right],$$

Gradient for the GLM log-likelihood

We can compute the gradient vector using the chain rule as follows

$$\begin{aligned}\frac{\partial \mathcal{L}_i}{\partial w_j} &= \frac{d\mathcal{L}_i}{d\theta_i}\frac{d\theta_i}{d\mu_i}\frac{d\mu_i}{d\eta_i}\frac{\partial \eta_i}{\partial w_j} \\ &= (y_i - A'(\theta_i))\frac{d\theta_i}{d\mu_i}\frac{d\mu_i}{d\eta_i}x_{i,j} \\ &= (y_i - \mu_i)\frac{d\theta_i}{d\mu_i}\frac{d\mu_i}{d\eta_i}x_{i,j}.\end{aligned}$$

This can be used inside a (stochastic) gradient descent procedure.

If we use a canonical link, $\theta_i = \eta_i$, this simplifies to

$$\mathbf{g}(\mathbf{w}) = \frac{1}{\sigma^2}\left[\sum_{i=1}^{N}(y_i - \mu_i)\mathbf{x}_i\right] = \frac{1}{\sigma^2}\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}), \text{ where } \boldsymbol{\mu} = [\mu_1, \cdots, \mu_N]^\top.$$

Hessian for the GLM log-likelihood

- For improved efficiency, we could use a second-order method.
- If we use a canonical link, the Hessian is given by

$$\mathbf{H}(\mathbf{w}) = -\frac{1}{\sigma^2}\sum_{i=1}^{N}\frac{d\mu_i}{d\theta_i}\mathbf{x}_i\mathbf{x}_i^\top = -\frac{1}{\sigma^2}\mathbf{X}^\top\boldsymbol{\Sigma}\mathbf{X},$$

where $\boldsymbol{\Sigma} = \mathrm{diag}\left(\frac{d\mu_1}{d\theta_1}, \cdots, \frac{d\mu_N}{d\theta_N}\right)$.

- This can be used inside the Iterative Reweighted Least Squares (IRLS) algorithm.

Iterative reweighted least squares (IRLS) algorithm

- The Iterative Reweighted Least Squares algorithm is a particular case of Newton's method.

- The updated parameters are obtained by iteratively solving a weighted least squares problem.

Least squares vs. weighted least squares

- A least squares (LS) problem refers to

$$LS(\mathbf{w}) = \min_{\mathbf{w}} \sum_{i=1}^{N} (y_i - \mathbf{w}^\top \mathbf{x}_i)^2,$$

for a dataset $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N = \{\mathbf{X}, \mathbf{y}\}$.

- It can be shown that the vector $\mathbf{w}$ that minimizes $LS(\mathbf{w})$ is given as

$$\mathbf{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

- A weighted least squares (WLS) problem refers to

$$WLS(\mathbf{w}) = \min_{\mathbf{w}} \sum_{i=1}^{N} r_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2,$$

for a dataset $\mathcal{D} = \{\mathbf{x}_i, y_i, r_i\}_{i=1}^N = \{\mathbf{X}, \mathbf{R}, \mathbf{y}\}$, with $\mathbf{R} = \mathrm{diag}(r_1, \ldots, r_N)$.

- It can be shown that the vector $\mathbf{w}$ that minimizes $WLS(\mathbf{w})$ is given as

$$\mathbf{w} = (\mathbf{X}^\top \mathbf{R} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{R} \mathbf{y}.$$

Iterative reweighted least squares problem

Newton's method for the log-likelihood of the GLM follows as

$$\begin{aligned}\mathbf{w}_{k+1} &= \mathbf{w}_k - \mathbf{H}_k^{-1}\mathbf{g}_k \\ &= \mathbf{w}_k + (\mathbf{X}^\top \boldsymbol{\Sigma}_k \mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}_k) \\ &= (\mathbf{X}^\top \boldsymbol{\Sigma}_k \mathbf{X})^{-1}[\mathbf{X}^\top \boldsymbol{\Sigma}_k \mathbf{X}\mathbf{w}_k + \mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}_k)] \\ &= (\mathbf{X}^\top \boldsymbol{\Sigma}_k \mathbf{X})^{-1}\mathbf{X}^\top \boldsymbol{\Sigma}_k \mathbf{z}_k,\end{aligned}$$

where $\mathbf{z}_k = \mathbf{X}\mathbf{w}_k + \boldsymbol{\Sigma}_k^{-1}(\mathbf{y} - \boldsymbol{\mu}_k)$ is known as the **working response**.

- Recall $WLS(\mathbf{w})$: $\mathbf{w} = (\mathbf{X}^\top \mathbf{R}\mathbf{X})^{-1}\mathbf{X}^\top \mathbf{R}\mathbf{y}$. At iteration k, the solution for $\mathbf{w}_{k+1}$ has a similar form to the solution for a weighted least squares problem, replacing $\mathbf{R}$ with $\boldsymbol{\Sigma}_k$, and $\mathbf{y}$ with $\mathbf{z}_k$.
- The name IRLS is due to at each iteration, we solve a weighted least squares problem, where the weight matrix $\boldsymbol{\Sigma}_k$ changes at each iteration.

Content

XX-XX-XX

- Linear regression, logistic regression, and Poisson regression
- The exponential family
- Generalized linear models (GLM)
- Iteratively reweighted least squares (IRLS)
- **GLM in Spark ML**

`GeneralizedLinearRegression()`

- It uses IRLS (Iterative Reweighted Least Squares) for optimization.
- It only allows ℓ_2 regularization.
- Spark currently only supports up to 4096 features through its `GeneralizedLinearRegression` interface.
- It will throw an exception if this constraint is exceeded.

GLMs available in PySpark

It includes the following families

Family	Response Type	Supported Links
Gaussian	Continuous	Identity*, Log, Inverse
Binomial	Binary	Logit*, Probit, CLogLog
Poisson	Count	Log*, Identity, Sqrt
Gamma	Continuous	Inverse*, Identity, Log
Tweedie	Zero-inflated continuous	Power link function

where * stands for canonical link.

For any random variable Z that obeys a Tweedie distribution, the variance $\mathrm{var}(Z)$ relates to the mean $\mathbb{E}(Z)$ by the power law: $\mathrm{var}(Z) = a\mathbb{E}(Z)^p$, where a and p are positive constants.

Parameters to adjust

- `family`: name of the family which describes the label distribution to be used in the model.
- `link`: name of link function which provides the relationship between the linear predictor and the mean of the distribution function.
- `maxIter`: max number of iterations.
- `regParam`: regularization parameter (≥ 0).

Particular cases: `LinearRegression()`

- If your **`family`** is '`gaussian`' and the **`link`** function is '`identity`', your GLM model is equivalent to linear regression.

- `LinearRegression()` is recommended instead of `GeneralizedLinearRegression()`.

- `LinearRegression()` allows for ℓ_1, ℓ_2, and elastic net regularization through L-BFGS or OWL-QN.

Particular cases: `LogisticRegression()`

- If your **`family`** is '`binomial`' and the **`link`** function is '`logit`', your GLM model is equivalent to logistic regression.

- `LogisticRegression()` is recommended instead of `GeneralizedLinearRegression()`.

- `LogisticRegression()` allows for ℓ_1, ℓ_2, and elastic net regularization through L-BFGS or OWL-QN.

References and recommended reading

Machine Learning: a Probabilistic Perspective

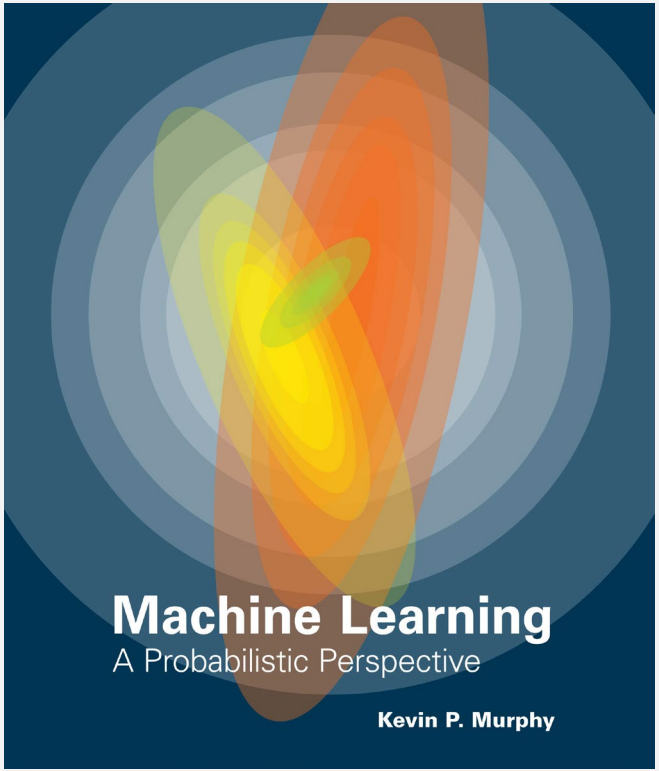
Chapter 9

The eBook is available via the University Library and can be accessed through the reading list on Blackboard.

Instructor's office hour:

Every Wednesday, 4:15 - 5:00 PM

@G25 Regent Court



References and recommended reading

Map-Reduce for Machine Learning on Multicore
by C-T Chu et al. (2006).

Map-Reduce for Machine Learning on Multicore

Cheng-Tao Chu *
chengtao@stanford.edu

Sang Kyun Kim *
skkim38@stanford.edu

Yi-An Lin *
ianl@stanford.edu

YuanYuan Yu *
yuanyuan@stanford.edu

Gary Bradski *†
garybradski@gmail

Andrew Y. Ng *
ang@cs.stanford.edu

Kunle Olukotun *
kunle@cs.stanford.edu

* CS. Department, Stanford University 353 Serra Mall,
Stanford University, Stanford CA 94305-9025.
† Rexee Inc.

Abstract

We are at the beginning of the multicore era. Computers will have increasingly many cores (processors), but there is still no good programming framework for these architectures, and thus no simple and unified way for machine learning to take advantage of the potential speed up. In this paper, we develop a broadly applicable parallel programming method, one that is easily applied to *many* different learning algorithms. Our work is in distinct contrast to the tradition in machine learning of designing (often ingenious) ways to speed up a *single* algorithm at a time. Specifically, we show that algorithms that fit the Statistical Query model [15] can be written in a certain "summation form," which allows them to be easily parallelized on multicore computers. We adapt Google's map-reduce [7] paradigm to demonstrate this parallel speed up technique on a variety of learning algorithms including locally weighted linear regression (LWLR), k-means, logistic regression (LR), naive Bayes (NB), SVM, ICA, PCA, gaussian discriminant analysis (GDA), EM, and backpropagation (NN). Our experimental results show basically linear speedup with an increasing number of processors.

References and recommended reading

Spark Python API Documentation for logistic regression: https://spark.apache.org/docs/4.1.0/api/python/reference/api/pyspark.ml.regression.GeneralizedLinearRegression.html

API Reference > MLlib (DataFrame-based) > GeneralizedLinearRegression

GeneralizedLinearRegression

class pyspark.ml.regression.GeneralizedLinearRegression(*, *labelCol='label', featuresCol='features', predictionCol='prediction', family='gaussian', link=None, fitIntercept=True, maxIter=25, tol=1e-06, regParam=0.0, weightCol=None, solver='irls', linkPredictionCol=None, variancePower=0.0, linkPower=None, offsetCol=None, aggregationDepth=2*) [source]

Generalized Linear Regression.

Fit a Generalized Linear Model specified by giving a symbolic description of the linear predictor (link function) and a description of the error distribution (family). It supports “gaussian”, “binomial”, “poisson”, “gamma” and “tweedie” as family. Valid link functions for each family is listed below. The first link function of each family is the default one.

- “gaussian” -> “identity”, “log”, “inverse”
- “binomial” -> “logit”, “probit”, “cloglog”
- “poisson” -> “log”, “identity”, “sqrt”
- “gamma” -> “inverse”, “identity”, “log”

Acknowledgment

Thanks to Dr. Mauricio Alvarez for his contribution to this session from 2017 to 2022